Report R11 — One rule space, two gates

What changes, and what cannot change, when the Hadamard is replaced by X

QCA Sector Analysis project (Tier 1)

2026-07-31 (R9 `1e.1` vs R10 `1x.1`, `obc0`, common window $N \leq 16$)

Abstract

R1–R9 analyse 256 brick-wall rules whose active gate is the Hadamard; R10 analyses the same 256 symbol tables with the gate changed to X. This report compares them. The two families turn out not to be independent: because applying the Hadamard to a site produces *both* values of the target bit, the flip is always one of the outcomes, so the X transition graph is a subgraph of the Hadamard one and the X sectors **refine** the Hadamard sectors — verified exhaustively for all 256 rules and over 2816 swept units, with zero exceptions. Three consequences organise everything else. The gate can only ever *split* sectors, never merge them, so no rule loses sector structure: of the 51 rules with an exponentially growing sector count under H, all 51 keep one under X, and 39 rules with no structure under H acquire it. The 81 rules with no `V` are literally the same circuit under both gates, which makes the two independent engines a cross-implementation test that they pass on all 916 shared units. And on the 16-rule unitary/reversible baseline — the only fully like-for-like comparison, since both families satisfy sectors = attractors there — the gate change is nearly a reflection: 9 of the 11 rules that have a single giant sector under H end up with no giant sector at all under X, 7 of them at the opposite extreme of the axis, with rule 204 the sole fixed point and rule 51 going from a single sector of 2^N states to 2^{N-1} two-cycles. Quantitatively almost nothing transfers: on the unitary baseline the Pearson correlation between the two gates' exponents is +0.17 in a and +0.03 in b. `obc0` only; `pbc` is held for both families.

Contents

1 What is being compared, and on what window

The rule encoding is shared exactly. A rule is a tuple $(r_{00}, r_{01}, r_{10}, r_{11})$ over $\{\mathtt{I}, \mathtt{V}, \mathtt{D}, \mathtt{E}\}$; the circuit applies the symbol selected by a site's two neighbours, brick-wall, even sublattice ascending and then odd, each reading the current state. Both engines compile the same step list from `core/cycle.py`. The single difference is what `V` means:

symbol	Hadamard family (R1–R9)	permutation family (R10)
`I`	identity	identity
`V`	Hadamard H	X
`D`	reset to $\|0\rangle$	reset to $\|0\rangle$
`E`	reset to $\|1\rangle$	reset to $\|1\rangle$

The window. R9 fits its bases on $N \le 16$ and R10 on $N \le 24$. A base fitted on a longer window is a different number, so every comparison below rebuilds the X-gate descriptors on **the same window as the Hadamard ones**, $N \le 16$, through the identical `scaling/sectors.py` machinery. Nothing is compared across windows. Where an $N = 24$ number is quoted it is labelled. One consequence is visible in the tables: rule 108's X-gate cycle length is classified *linear* on $N \le 16$ and *exponential* on $N \le 24$, which is a statement about how much data the classifier has, not about the rule.

What is not compared. The Hadamard family carries objects that need interference between distinct branches of a basis state — destructive cancellation in succ, the R7 visibility theorem's sharp cases. It does *not* have a monopoly on protected coherence: R10 §9 finds decoherence-free subspaces of exponential dimension throughout the permutation family, and for the 81 V-free rules they are literally the same subspaces, since the channel is the same. Conversely the permutation family's transient structure has no Hadamard counterpart worth plotting. The comparison is restricted to the two coordinates both families define: the sector count $n_{\mathrm{wcc}} \sim a^N$ and the largest sector $D_{\max} \sim b^N$, plus the growth *class* of each.

2 The two families are nested

The comparison is not between two unrelated data sets. There is an exact containment, and it is elementary.

Theorem 1 (refinement)**.** *For every rule, every N and both boundary conventions, $\mathrm{succ}_X(x) \in \mathrm{succ}_H(x)$ for every basis state x. Hence the X transition graph is a subgraph of the Hadamard transition graph on the same vertex set, and the X sectors refine the Hadamard sectors: every X-component lies inside a single H-component, and every Hadamard sector is a disjoint union of X sectors.*

Proof. Consider one elementary step. At `I`, `D`, `E` sites the two families do the same thing. At a `V` site the Hadamard maps $|b\rangle$ to $(|0\rangle \pm |1\rangle)/\sqrt{2}$, so the measured or Kraus-resolved successor set contains *both* values of the target bit; the X action, which sets it to $1 - b$, is one of them. The circuit is a composition of such steps, and choosing the flip at every `V` site along the sweep is one branch of the Hadamard successor set. So each X edge is an H edge. Union-find over a subgraph produces a refinement. □

Corollary 1. *$n^X_{\mathrm{wcc}}(N) \geq n^H_{\mathrm{wcc}}(N)$ and $D^X_{\max}(N) \leq D^H_{\max}(N)$, pointwise in N; asymptotically $a_X \geq a_H$ and $b_X \leq b_H$.*

Corollary 2. *A rule with no* `V` *never invokes the gate, so its two circuits are the* same map. *There are* 81 *such rules (*204 *and the* 80 *V-free ones), and for them the refinement is an equality.*

All of this is verified rather than assumed:

check	window	holds	unit
$\mathrm{succ}_X(x) \in \mathrm{succ}_H(x)$	$N = 8,\ N = 9$	256/256	rules, all states
X sectors refine H sectors	$N = 8,\ N = 9$	256/256	rules, exhaustive
$n^X_{\mathrm{wcc}} \geq n^H_{\mathrm{wcc}}$	$N \leq 16$	2816/2816	units
$D^X_{\max} \leq D^H_{\max}$	$N \leq 16$	2816/2816	units
V-free rules identical in both stores	all N	916/916	units, 81 rules

The last row is worth its own sentence. The Hadamard sectors come from a set-valued successor relation streamed through union-find; the X-gate sectors come from a numpy step table and a three-colour walk. The two share no code below `core/cycle.py::_compile`. On the 81 rules where they must agree they agree exactly — the full sector-size multiset, at every N where both were run. That is the strongest cross-implementation check either report has.

Reading the theorem physically. The Hadamard's extra edge is the non-flip branch, and it is exactly what *glues* X sectors together. So one may think of the Hadamard sector structure as the X sector structure after gluing: superposition is a coarsening operation on the sector lattice. That is the reverse of the intuition one might start with — that a superposing gate would *create* structure — and it explains the census in the next section without any further input.

3 No rule loses its sector structure

Corollary 1 says the sector count cannot fall. The growth-class cross-tabulation makes the consequence concrete:

	under X				
under H	constant	linear	exponential	irregular	all
constant	134	6	28	5	173
polynomial	0	20	10	0	30
exponential	0	0	51	0	51
irregular	0	0	0	2	2
all	134	26	89	7	256

Every entry below the diagonal is zero, as it must be. Reading the rows:

- **All 51 rules with an exponential sector count under the Hadamard keep one under X.** Hilbert-space fragmentation, measured by the growth class of the sector count, is not a property of the Hadamard: it survives replacing the superposing gate by a permutation.
- 39 of the 173 rules with a *constant* sector count under H — no structure at all, one or two sectors covering the space — acquire structure under X: 28 exponential, 6 linear, 5 irregular. The gate change can only add.
- 10 of the 30 rules with a linear sector count under H become exponential; the other 20 stay linear.

The reverse reading is the one that matters for the Hadamard reports: if a rule has no sector structure under X, it has none under H either. The 134 rules in the top-left cell are structureless under both gates, and that is a property of their symbol table rather than of the dynamics.

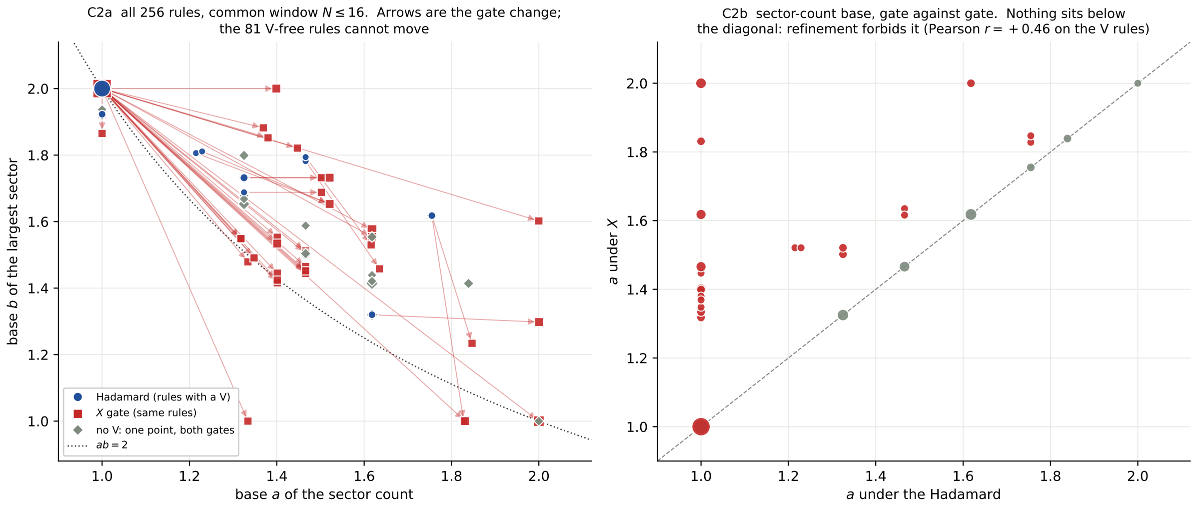

Figure 1: **C2, all 256 rules on a common window.** Left: the sector plane, Hadamard point (circle) joined to X point (square) for each rule that has a `V`; the 81 V-free rules are drawn as diamonds, one point each, because the gate cannot move them (Corollary 2). Almost every Hadamard point sits at $(1, 2)$ — one giant sector — and the gate change fans them out to the right. Right: the sector-count base under one gate against the other. Nothing lies below the diagonal, which is Corollary 1 drawn.

4 The unitary / reversible baseline

The 16 rules built from `I` and `V` alone are the cleanest comparison available, for a specific reason: on the Hadamard side they are unitary, so sectors and monitored attractors coincide (R9's check A2), and on the X side they are reversible, so sectors and cycles coincide (R10's F1). The same object is being measured in both families, and the comparison cannot be confounded by the transient structure that dominates everywhere else.

		Hadamard		X		sector-count class	
rule	tuple	a	b	a	b	H	X
51	`VVVV`	1.0000	2.0000	2.0000	1.0000	constant	exponential
54	`IVVV`	1.0000	2.0000	2.0000	1.0000	constant	exponential
57	`VIVV`	1.0000	2.0000	1.3994	2.0000	constant	exponential
60	`IIVV`	1.0000	2.0000	2.0000	1.0000	polynomial	exponential
99	`VVIV`	1.0000	2.0000	1.3994	2.0000	constant	exponential
102	`IVIV`	1.0000	2.0000	2.0000	1.0000	polynomial	exponential
105	`VIIV`	1.0000	2.0000	2.0000	1.0000	polynomial	exponential
108	`IIIV`	1.7549	1.6180	1.8278	1.0000	exponential	exponential
147	`VVVI`	1.0000	2.0000	2.0000	1.0000	constant	exponential
150	`IVVI`	1.0000	2.0000	2.0000	1.0000	polynomial	exponential
153	`VIVI`	1.0000	2.0000	1.8311	1.0000	constant	exponential
156	`IIVI`	1.6180	1.3195	2.0000	1.2983	exponential	exponential
195	`VVII`	1.0000	2.0000	1.8311	1.0000	constant	exponential
198	`IVII`	1.6180	1.3195	2.0000	1.2983	exponential	exponential
201	`VIII`	1.7549	1.6180	1.8470	1.2339	exponential	exponential
204	`IIII`	2.0000	1.0000	2.0000	1.0000	exponential	exponential

Three things stand out, and none of them is a small effect.

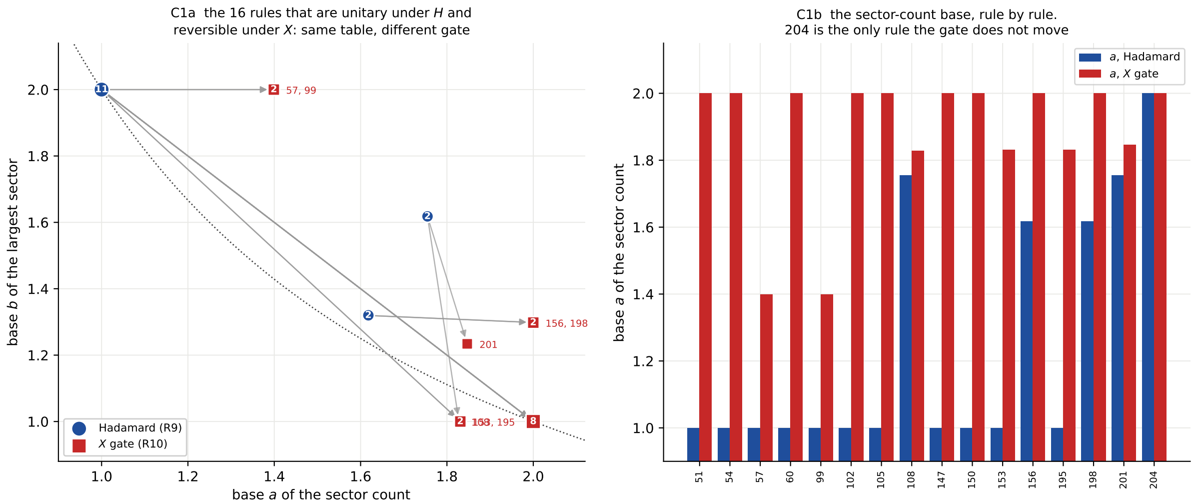

Figure 2: **C1, the baseline.** Left: each of the 16 rules under both gates, joined by the move; marker area counts rules sharing a cell. Right: the sector-count base rule by rule. Rule 204 is the only one the gate leaves where it was.

The gate nearly reflects the family. Eleven of the 16 sit at $(a, b) = (1, 2)$ under the Hadamard — a single sector of size 2^N, no fragmentation whatever. Under X, nine of those eleven end at $b = 1$: the largest sector is no longer a fixed fraction of the space at all. Seven go to $(2, 1)$ exactly — exponentially many tiny orbits, maximal fragmentation — and 153 and 195 to $(1.83, 1)$. Only 57 and 99 keep $b = 2$, moving left instead of down. The extreme case is rule 51 (`VVVV`). Under the Hadamard it is the most ergodic rule in the family, one sector covering the whole basis. Under X it is $x \mapsto \bar{x}$, so the space splits into 2^{N-1} two-cycles. Same symbol table, opposite ends of the sector axis, and Theorem 1 says which way round it must be: the Hadamard picture is the X picture after gluing.

Only 204 is fixed. Rule 204 is `IIII`, the identity, which contains no `V` and therefore cannot move (Corollary 2). It is the only one of the 16 with that property, and correspondingly the only rule whose a is identical in the two columns.

The four fragmented rules stay fragmented, with different constants. 156, 198, 108, 201 are the four rules whose sector count grows exponentially under the Hadamard, and all four still do under X — but the constants have no relation:

	Hadamard	X
156, 198	$(\varphi,\ 4^{1/5})$	$(2,\ 1.298)$
108, 201	$(1.7549,\ \varphi)$	$(1.828,\ 2)$, $(1.847,\ 1.234)$

The Hadamard values are exact algebraic numbers with derivations behind them (R2 §3's room-packing saddle point for $4^{1/5}$; the golden ratio and the root of $x^3 = 2x^2 - x + 1$ from integer recurrences). The X-gate values are not the same numbers and are not obviously algebraic. Fragmentation transfers; the *grammar* producing it does not.

And the correlation is nil.

subset	coordinate	rules	Pearson r	mean $\|\Delta\|$	unchanged
all 256	a (sector count)	249	+0.642	0.100	199
all 256	b (largest sector)	251	+0.561	0.095	202
the 175 with a `V`	a (sector count)	170	+0.465	0.146	120
the 175 with a `V`	b (largest sector)	172	+0.334	0.138	127
the 16 unitary/reversible	a (sector count)	16	+0.171	0.649	1
the 16 unitary/reversible	b (largest sector)	16	+0.034	0.628	3
the 240 others	a (sector count)	233	+0.759	0.062	198
the 240 others	b (largest sector)	235	+0.698	0.058	199

On the 16 baseline rules the Pearson correlation between the two gates is +0.17 in a and +0.03 in b, with mean absolute differences of 0.65 and 0.63 on axes of total width 1. Read the "all 256" row with care: its apparent $r = +0.64$ is inflated by the 81 rules that are the same circuit under both gates and therefore correlate perfectly by construction. Restricted to the 175 rules that actually contain a `V`, r falls to +0.47 and +0.33. The gate does not preserve exponents.

5 R9's headline sets under the other gate

R9 singled out three sets of rules. Each is asked here what it does under X.

		H: sector count		X: sector count		X cycle
rule	tuple	class	at $N = 16$	class	at $N = 16$	class
open-system fragmented (R9 §6.3)						
156	`IIVI`	exponential	2 584	exponential	7 744	exponential
198	`IVII`	exponential	2 584	exponential	7 744	exponential
140	`IIDI`	exponential	2 584	exponential	2 584	constant
206	`IEII`	exponential	2 584	exponential	2 584	constant
196	`IDII`	exponential	2 584	exponential	2 584	constant
220	`IIEI`	exponential	2 584	exponential	2 584	constant
108	`IIIV`	exponential	10 252	exponential	16 560	polynomial
201	`VIII`	exponential	5 842	exponential	12 076	exponential
strictly linear sector count (R9 §6.4)						
44	`IIDV`	polynomial	17	exponential	595	constant
100	`IDIV`	polynomial	19	exponential	595	constant
110	`IEIV`	polynomial	17	exponential	149	polynomial
124	`IIEV`	polynomial	17	exponential	149	polynomial
188	`IIVE`	polynomial	17	polynomial	17	polynomial
230	`IVIE`	polynomial	17	polynomial	17	polynomial
pinned frontier (R9 §6.5)						
44	`IIDV`	polynomial	17	exponential	595	constant
60	`IIVV`	polynomial	17	exponential	4 116	constant
100	`IDIV`	polynomial	19	exponential	595	constant
102	`IVIV`	polynomial	17	exponential	4 116	constant
110	`IEIV`	polynomial	17	exponential	149	polynomial
124	`IIEV`	polynomial	17	exponential	149	polynomial
188	`IIVE`	polynomial	17	polynomial	17	polynomial
230	`IVIE`	polynomial	17	polynomial	17	polynomial

5.1 The open-system-fragmented rules

R9's central physical claim is that eight V+reset rules keep an exponentially growing sector count, and that because a sector is an enclosure — span of a weak component is invariant under every Kraus operator — this is fragmentation of an *open* quantum system rather than of a unitary circuit with noise added afterwards.

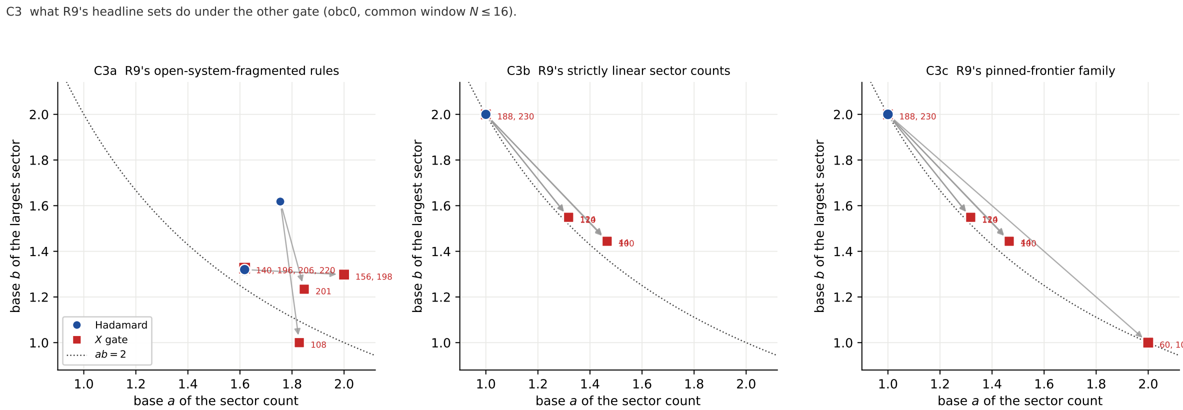

Figure 3: **C3, the three sets.** Hadamard (circle) to X (square) in the sector plane, common window. Labels list the rules sharing a destination cell.

All eight remain exponential under X, and four of them do something stronger: 140, 196, 206, 220 have *exactly* the same sector count as under the Hadamard, 2584 at $N = 16$, because they are V-free and hence the same circuit (Corollary 2). Their appearance in R9's list is therefore not a statement about the Hadamard at all: those four rules are classical, and their fragmentation is a classical fact that R9 measured with a quantum engine. The other four — 156, 198, 108, 201 — do contain a `V`, and for them the X count is strictly larger (7744 against 2584; 16560 against 10252; 12076 against 5842), exactly as Corollary 1 requires.

This is a sharpening of R9 rather than a contradiction of it. The claim "these rules fragment as open quantum systems" stands; what R11 adds is that for half of them the Hadamard was never doing any work.

5.2 The linear-sector-count rules and the pinned frontier

R9 §6.4 identified six rules — 44, 100, 110, 124, 188, 230 — whose sector count is strictly linear, $N + 1$ of them, carried by a charge that is the position of the extremal excitation; §6.5 showed the charge is an `obc0` artefact, collapsing to two sectors on the ring. Under X the six split three ways:

- 188 (`IIVE`) and 230 (`IVIE`) keep *exactly* 17 sectors at $N = 16$, the same linear count as under the Hadamard, though they contain a `V` and are therefore not forced to. For these two the pinned-frontier charge is gate-independent.
- 110 and 124 become exponential but only just, 149 sectors at $N = 16$ against 17.
- 44 and 100 jump to 595.

The frontier family of §6.5 adds 60 and 102, and those two go all the way to 4116 sectors — from a linear count to $a = 2$, the maximum. So the pinned frontier is a Hadamard-specific structure for six of the eight rules that carry it, and survives the gate change for two. Combined with R9's finding that it does not survive the ring, the charge is doubly fragile: it depends on the boundary condition *and*, mostly, on the gate. That is a good reason not to build a physical story on it, and it is the sort of thing the comparison is for.

6 Reading all of this as Hilbert-space fragmentation

The two coordinates used throughout are exactly the two numbers the HSF literature uses to classify a fragmented model, so the map is a phase diagram and it is worth saying so explicitly.

6.1 The (a, b) plane is the fragmentation phase diagram

A model is *fragmented* when the Hilbert space — or a symmetry sector of it — splits into exponentially many dynamically disconnected Krylov subspaces. With $n_{\mathrm{wcc}} \sim a^N$ subspaces and a largest one of dimension $D_{\max} \sim b^N$:

$a = 1$, $b = 2$	**unfragmented**	one Krylov sector carries the space
$a > 1$, $b = 2$	**weak**	exponentially many, largest still $\sim 2^N$
$a > 1$, $1 < b < 2$	**strong**	largest sector an exponentially vanishing fraction
$b = 1$	**shattered**	every sector sub-exponential

The exclusion curve $ab \geq 2$ is then the statement that one cannot have few sectors *and* small ones, which is why the weak/strong boundary at $b = 2$ and the shattered edge at $b = 1$ are the two edges of the diagram rather than independent axes. The corner $a = 1$, $b < 2$ is forbidden asymptotically, and every rule the fitter puts there has a sub-exponential sector count that the base cannot express (R9 §5); those are labelled "degenerate" below and are a fitting category, not a phase.

	under X						
under H	unfragmented	weak	strong	shattered	degenerate	irregular	all
unfragmented	152	2	26	10	1	5	196
strong	0	0	49	1	0	0	50
shattered	0	0	0	1	0	0	1
degenerate	0	0	0	0	7	0	7
irregular	0	0	0	0	0	2	2
all	152	2	75	12	8	7	256

Refinement (Theorem 1) forbids the lower-left of this table too: a gate change to X can only move a rule *towards* more fragmentation. And it does. Under the Hadamard 50 of 256 rules are strongly fragmented; under X, 75 are, plus 12 shattered. Not one rule becomes less fragmented. The two *weak* entries are 57 and 99, the only rules anywhere in either family with exponentially many sectors and a largest sector still of full exponential size.

6.2 Rules 156 and 198: a configuration space, or a bundle of orbits

The clearest single comparison in the report. Rules 156 (`IIVI`) and 198 (`IVII`) are strongly fragmented under *both* gates — and mean something completely different by it.

Under the Hadamard: a constrained configuration space. R2 §3 derives the structure. Domain walls are frozen; the chain decomposes into "rooms" between them; a room of length ℓ contributes $\ell + 1$ basis states and consumes $\ell + 2$ sites. Counting the wall patterns gives $n_{\mathrm{wcc}} \sim \varphi^N$ and maximising the room packing gives $D_{\max} \sim (4^{1/5})^N$, the saddle point of $b(\ell) = (\ell + 1)^{1/(\ell+2)}$. So a sector is labelled by a *static* charge — the wall pattern — and inside it the accessible states form an exponentially large constrained configuration space, on which the Hadamard acts as a genuine quantum dynamics.

Under X: the same sectors, cut into orbits by extra charges. The sectors cannot be finer than the Hadamard ones by anything other than refinement, and indeed the wall pattern is still conserved. But the X dynamics is a permutation, so each Hadamard sector decomposes further — and it does so with a very particular arithmetic. At $N = 14$ rule 156's 987 Hadamard sectors split into 2280 X orbits, and **every sector splits into orbits that are all the same length**:

$$\dim(H \text{ sector}) \;=\; (\text{number of } X \text{ orbits}) \times (\text{orbit length}).$$

dim of the H sector	X orbits	orbit length	sectors
2	1	2	28
3	1	3	21
4	1	4	21
4	2	2	35
6	1	6	85
8	2	4	40
8	4	2	35
9	3	3	20
10	1	10	46
12	1	12	43
12	2	6	65
14	1	14	21
15	1	15	20
16	4	4	55
16	8	2	6
18	3	6	65
20	1	20	20
24	2	12	38
24	4	6	39
27	9	3	5
30	1	30	40
32	8	4	19
32	16	2	1
35	1	35	2
36	1	36	2
40	1	40	2
42	1	42	8
48	4	12	21
54	9	6	7
60	1	60	6
64	16	4	1

Read the arithmetic. A sector of dimension 16 appears as 2×8, 4×4 or 8×2; one of dimension 36 as 3×12 or 6×6; one of dimension 64 as 16×4. The orbit lengths that occur are 1 through 15, then 18, 20, 21, 22, 24, 28, 30, 35, 36, 40, 42 and 60 — and $35 = \mathrm{lcm}(5,7)$, $42 = \mathrm{lcm}(6,7)$, $60 = \mathrm{lcm}(4,15)$. At $N = 24$ the longest orbit in the whole rule is 420, and the sequence leading there is $105, 140, 140, 210, 210, 420, 420, 420$, i.e. $\mathrm{lcm}(3,5,7)$, $\mathrm{lcm}(4,5,7)$, $\mathrm{lcm}(2,3,5,7)$, $\mathrm{lcm}(3,4,5,7)$.

That is the signature of *independent periodic subsystems.* The natural reading, consistent with every number above though not derived here: inside a fixed wall pattern each room carries its own cyclic degree of freedom of some period p_r; the X dynamics advances all of them together; so the sector dimension is $\prod_r p_r$, an orbit has length $\mathrm{lcm}_r(p_r)$, and the number of orbits is $\prod_r p_r/\mathrm{lcm}_r(p_r)$ — the *relative phases* between rooms. Those relative phases are exactly the extra conserved charges that the X gate has and the Hadamard has not: the Hadamard superposes the room states and destroys the phase label, gluing the orbits back into one sector.

The property is specific, not generic: it holds for 156 and 198 at every N checked, and fails for 108, 201, 54, 150 and 105, whose sectors contain orbits of differing length.

6.3 Why the Hadamard result is the interesting one

Three reasons, in increasing order of weight.

The sectors have room in them. Strong fragmentation is only a statement worth making if a sector is big enough to have its own thermodynamics. Rule 156 under the Hadamard has φ^N sectors of size up to $(4^{1/5})^N$ — both exponential, product $2.135 > 2$ — so each Krylov sector is an exponentially large quantum system in its own right, and the questions HSF is asked (does a sector thermalise? what is the level statistics inside it? what does an initial state relax to?) all have content. Under X the same rule has $\sim 2^N/\text{poly}$ orbits whose typical length is $O(N)$: exponentially many "sectors", each of which is a single trajectory.

It is visible in an entanglement measurement. Under the Hadamard an initial computational basis state spreads over its sector, and R8 §10 measured exactly this for rule 150 at $N = 11$: the saturation value of the entanglement entropy clusters by sector, each cluster sitting near the Page value of its own sector dimension. The fragmentation is therefore an observable statement about a quantum state. Under X a basis state remains a basis state for all time, so $S(t) \equiv 0$ and there is nothing to measure — the fragmentation is invisible to any entanglement probe applied to basis states.

Every *sector* label of the X circuit is classical. A permutation of the basis conserves the indicator function of each of its orbits, so a fragmented automaton has an exponentially large commutant for free. That is not a discovery about the model; it is the statement that a deterministic map has trajectories. What R9 finds is the same object in a setting where the sector label is still a classical function on configurations but the dynamics inside is not — and where, by the diagonal-commutant identity of R8 §10.3, the sector projectors are precisely the diagonal part of the commutant, i.e. everything a classical partition can see and nothing more. The qualifier matters: the *off*-diagonal part of the commutant of an X-gate rule with resets is not trivial either, and R10 §9 measures it.

6.4 Why the X-gate and the V-free rules are the boring end

The same three points, run in reverse, and they are worth stating plainly because the numbers in §3 look impressive on their own.

Under X the 256 rules split into two kinds and neither supports an interesting fragmentation claim. For the 16 reversible rules, sectors *are* orbits, so the decomposition exists for every bijection whatsoever and contains no information about locality, constraints or interactions. For the other 240, the sectors are basins of attraction: they can be exponentially large, but the long-time state is a cycle of median length 1 and a cyclic fraction of 10^{-7}, so a sector is an exponentially large funnel into a point. Neither is Hilbert-space fragmentation in the sense the term is used for constrained quantum models.

The 81 V-free rules make the point sharpest, because they are the same circuit under both gates (Corollary 2). Whatever *fragmentation* they show is a property of a classical cellular automaton that a quantum engine happened to compute. Their *coherence* structure is another matter: rule 232 (`DIIE`) is V-free and has a decoherence-free subspace of dimension $\sim \varphi^N$, confirmed against the exact Cesaro rank ($\dim \mathrm{Fix}(\Phi) = n_{\mathrm{rec}}^2$, R10 §9.4). A classical-population circuit can protect an exponentially large quantum subspace, and that is not a classical statement. Four of R9's eight "fragmentation in an open quantum system" rules — 140, 196, 206, 220 — are in this set, so for those four the claim should be read as classical fragmentation surviving a reset, with the Hadamard playing no part. The other four (156, 198, 108, 201) do use the gate, and for them the claim stands as stated.

6.5 Automaton circuits: why the community calls them quantum

The natural objection to R10 is that it studies a classical cellular automaton and calls it a QCA. The objection is half right, and the half that is wrong is instructive.

Unitarity is not quantumness. A permutation matrix is unitary, so a reversible CA *is* a quantum cellular automaton under the structural definition — discrete time, strict locality, a causal cone, a unitary on $(\mathbb{C}^2)^{\otimes N}$. Nothing is being smuggled. What is true is that unitarity alone buys none of the phenomena the word "quantum" is meant to signal.

The quantum content is in the initial state, not the gate. A permutation circuit maps basis states to basis states, but it is linear, and on a *superposition* it does not act classically at all. Take $|+\rangle^{\otimes N}$ or any product state off the computational axis: an automaton circuit will entangle it, and can scramble it as fast as a generic circuit. This is the reason the class is studied. It sits in the solvable corner — classically simulable trajectory by trajectory, so accessible at sizes no exact diagonalisation reaches, while still supporting entanglement growth, operator spreading and hydrodynamics that are genuinely quantum questions. Our rule 1 result in R10 §5 is of this type: its recurrent set is exactly the golden-mean shift, which is a statement about a classical constraint, but it fixes the long-time support of any quantum state prepared in that sector.

Where the class comes from. Two lines of work are the relevant context, and the connection is concrete rather than thematic.

Rule 54. The most-studied exactly solvable QCA is the Rule 54 automaton (studied by, among others, Prosen and collaborators, and by Klobas and Bertini for its exact steady states and entanglement dynamics). It is a deterministic reversible cellular automaton whose local update, in the brick-wall formulation, is $x_i \mapsto x_i \oplus (x_{i-1} \vee x_{i+1})$ — and that is exactly our `IVVV`, which our Wolfram numbering calls rule 54. More generally, for all 16 reversible rules the local update is $x_i \mapsto x_i \oplus f(x_{i-1}, x_{i+1})$ with f one of the 16 boolean functions of two arguments, and the induced elementary CA is the ECA of the *same Wolfram number* — we checked all 16. So the reversible X-gate family is precisely the set of ECAs that are affine in the centre cell, brick-walled, and Rule 54 is one of its members. R10's structural results therefore apply to it directly: X_{54} has $2^N/\text{poly}$ orbits of length $52, 55, 58, 61, 64, 67$ at $N = 19 \dots 24$, growing by exactly 3 per site, which is the soliton picture on an open chain seen through the sector axis.

Quantum automaton circuits. The other line uses exactly this class as a tool: unitaries that permute the computational basis, deployed to study fragmentation, freezing and anomalously slow transport at sizes where the entanglement can still be computed — work associated with Iadecola and collaborators, and with the automaton-circuit studies of subdiffusion and subsystem symmetries. The methodological bargain is explicit there: classical simulability in exchange for a restricted class of dynamics, with the quantum questions asked of superposed initial states.

What this report adds to that. A complete 256-rule census of the sector and orbit structure of the class, and one structural statement that the automaton literature and the constrained-quantum-model literature do not usually put side by side: the automaton sectors *refine* the Hadamard ones (Theorem 1). An automaton model is therefore a lower bound on the fragmentation of its superposing counterpart, never an upper bound, and any fragmentation found in an automaton circuit should be checked against the question of whether it survives gluing — which for 39 of our 256 rules it does not.

A caveat on the attributions. The two paragraphs above characterise the literature from memory, without the papers to hand; the identification of X_{54} with Rule 54 is our own check and is exact, but the names and the framing attached to the two research lines should be verified against the sources before this section is used anywhere outside the project.

7 What each family says that the other cannot

The comparison is not symmetric, and the asymmetries are informative.

Only the Hadamard family has branch interference — which is less than it sounds. Destructive cancellation between two branches of a basis state needs a successor set with more than one element, and under X there is only ever one, so that specific mechanism is absent. It does not follow that the permutation family is decoherence-free-trivial, and an earlier draft of both reports said that it did. A reset has two Kraus operators which agree on populations and differ on coherences, so the functional graph — a statement about populations alone — cannot see decoherence at all. R10 §9 computes what it misses: at $N = 12$, 127 of the 240 irreversible X rules carry a decoherence-free subspace of dimension > 1, and the dimensions are exponential, with rule 1's equal to the golden-mean shift (F_{N+2}), rules 76 and 73's to the no-`111` shift (tribonacci), and rule 22's to the Lucas numbers L_{N+1}. The X-gate family is therefore a classical control *for populations only.* That is still the useful reading — §3 shows the sector axis is almost entirely graph combinatorics — but the coherence axis is not classical at all.

Only the permutation family has a clean recurrent structure. With out-degree one, $n_{\text{rec}} = n_{\text{wcc}}$ identically and every basin is a sector, so questions about attractors have exact answers (R10 §§5–8). In the Hadamard family the same questions are genuinely harder — rule 22 has three attractors inside two sectors — which is why R9 could assert A4 in one direction only. The permutation family is where the attractor-level statements can be proved.

The exclusion curve belongs to both. $ab \geq 2$ is a theorem about partitions and applies wherever sectors partition the basis, so it binds both families identically, and neither produced a violation. It does *not* apply to cycles, and R10 §5 makes that precise: the recurrent set is not the whole space, so the analogous constraint degenerates.

8 Open items

1. `pbc`**.** Held for both families. The refinement theorem is proved without reference to the boundary, so it should carry over verbatim, and it will be worth checking that the ring does not break the 81-rule identity.
2. **The gluing map.** Theorem 1 says each Hadamard sector is a union of X sectors, but nothing here counts them: the natural next observable is the number of X sectors per Hadamard sector, whose growth would measure exactly how much the non-flip branch glues.
3. **A third gate.** X and H are the two extremes — a permutation and a maximally superposing gate. A partial rotation would interpolate, and the refinement theorem suggests the sector lattice would interpolate monotonically with it.
4. **The four classical rules in R9's open-fragmented list.** 140, 196, 206, 220 deserve a purely combinatorial derivation of their 2584 sectors; nothing quantum is involved and the count is Fibonacci.

9 Reproduce

```
cd code
python -m qca_fragmentation.permutation.compare --bc obc0
python -m qca_fragmentation.permutation.compare_tables --bc obc0
python -m qca_fragmentation.permutation.compare_figures --bc obc0 --rebuild
python -m pytest qca_fragmentation/tests/test_gate_compare.py -q
```

Code: `permutation/compare.py` (the refinement checks, the common-window descriptors, the correlations), `permutation/compare_tables.py`, `permutation/compare_figures.py`. Data: `results_wcc/` and `results_xgate/`, `analytics/gate_compare_obc0.json`.